

OPARINO, Russian Federation

WMO: 270830

Lat: 59.850N

Lon: 48.283E

Elev: 589

StdP: 14.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-18.7	-29.2	1.1	-24.6	-23.2	1.6	-18.6	14.2	21.5	12.8	20.4	1.8	10	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	82.6	67.4	79.2	65.6	75.9	63.7	69.4	79.4	67.6	76.2	65.8	73.3	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	98.0	73.8	64.4	92.5	71.8	62.6	86.7	70.2	33.8	79.9	32.3	76.5	30.9	73.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.8	10.3	DB	-31.5	87.2	8.0	3.1	-37.2	89.4	-41.9	91.2	-46.4	93.0	-52.2	95.2	(4)
(5)			WB	-31.7	72.7	7.9	3.1	-37.4	74.9	-42.0	76.8	-46.4	78.5	-52.2	80.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	9.1	11.7	23.6	37.4	49.7	58.2	63.1	58.0	48.4	36.5	23.3	13.8		
	DBStd	21.23	14.21	13.55	9.08	8.79	9.50	7.93	6.84	7.26	7.26	7.83	11.06	14.35		
	HDD50	6165	1269	1071	819	387	127	17	2	12	113	425	801	1122		
	HDD65	10626	1734	1491	1284	828	479	230	119	240	498	885	1251	1587		
	CDD50	1130	0	0	0	9	118	264	407	261	66	5	0	0		
	CDD65	116	0	0	0	0	6	27	59	23	1	0	0	0		
	CDH74	1115	0	0	0	3	98	272	541	194	7	0	0	0		
	CDH80	234	0	0	0	0	15	56	115	48	0	0	0	0		
	WSAvg	4.5	4.6	4.8	5.0	4.6	4.7	4.2	3.5	3.9	4.3	5.0	4.9	5.0		
	PrecAvg	26.3	1.6	1.3	1.5	1.3	2.2	3.3	3.0	3.3	2.3	2.4	2.1	1.9		
	PrecMax	34.0	2.6	3.1	3.1	2.5	5.4	6.1	6.9	7.2	4.8	4.3	4.2	3.3		
	PrecMin	19.8	0.7	0.4	0.2	0.1	0.6	0.7	0.4	1.0	0.6	0.9	0.6	1.2		
	PrecStd	3.7	0.4	0.7	0.7	0.6	1.0	1.5	1.7	1.2	1.1	1.0	0.8	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	35.4	48.0	69.8	82.0	86.4	87.5	86.0	73.7	62.2	46.4	38.4		
		MCWB	32.2	33.2	39.1	54.3	63.3	69.4	69.8	69.9	60.8	50.1	44.7	37.7		
	2%	DB	32.3	33.3	42.2	61.7	76.3	80.7	83.3	79.2	67.9	54.8	40.6	34.8		
		MCWB	31.5	32.2	35.6	48.7	59.2	65.8	68.4	67.3	58.8	49.0	39.1	34.1		
	5%	DB	30.3	31.8	38.1	56.1	71.3	76.7	79.9	74.4	63.8	50.7	37.4	32.8		
		MCWB	29.5	30.9	33.2	44.9	56.2	63.5	66.8	64.0	56.5	47.0	36.4	32.1		
	10%	DB	27.4	29.5	35.2	51.6	67.0	73.0	76.7	70.1	59.9	47.4	34.9	31.1		
		MCWB	26.5	28.3	31.9	42.6	54.5	61.9	65.2	62.0	54.7	45.0	34.0	30.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	33.9	40.0	54.9	65.6	71.4	71.8	70.9	62.6	53.8	44.8	37.8		
		MCDB	33.3	34.5	46.4	69.2	78.2	83.3	82.6	82.6	69.4	59.7	46.2	38.2		
	2%	WB	31.7	32.5	36.6	49.6	61.5	68.3	69.8	68.2	59.9	49.9	39.5	34.2		
		MCDB	32.3	33.3	40.5	60.0	72.2	77.4	79.6	76.9	66.0	53.5	40.3	34.5		
	5%	WB	29.5	30.9	34.2	46.3	58.5	65.9	68.1	65.9	57.6	47.4	36.6	32.1		
		MCDB	30.3	31.9	37.0	54.2	68.0	73.0	77.3	72.4	62.8	50.2	37.3	32.6		
	10%	WB	26.5	28.3	32.4	43.3	55.7	63.3	66.5	63.3	55.2	45.0	34.2	30.4		
		MCDB	27.3	29.3	34.8	50.6	65.2	70.8	74.9	68.6	59.3	47.4	34.8	31.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	13.0	14.4	16.8	19.3	17.9	17.8	15.6	13.5	8.2	7.3	9.1		
		MCDBR	9.1	9.9	17.8	25.5	26.7	23.7	23.6	21.7	20.7	14.6	8.1	6.0		
		MCWBR	9.3	9.5	12.4	14.7	12.8	11.3	10.5	10.2	11.9	9.8	7.7	6.2		
	5% WB	MCDBR	9.1	9.6	13.5	22.7	23.2	20.5	20.8	19.9	17.6	13.1	7.6	6.3		
		MCWBR	9.3	9.5	10.1	14.1	12.4	10.8	10.3	10.4	11.0	9.6	7.4	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.303	0.340	0.376	0.361	0.359	0.391	0.384	0.348	0.325	0.286	0.246		
	taud		2.034	2.065	2.046	2.158	2.418	2.449	2.363	2.364	2.446	2.372	2.148	1.987		
	Ebn at Noon		153	213	243	259	275	277	264	256	247	212	147	111		
	Edh at Noon		15	26	36	39	33	32	34	32	25	19	14	10		
(44) All-Sky Solar Radiation	RadAvg		112	332	718	1185	1575	1670	1689	1215	694	295	102	58		
	RadStd		17	82	110	149	155	142	165	126	93	40	17	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.06	N/A	N/A	N/A	N/A	+1.13	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page